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COURSE : CSA0927 – PROGRAMMING IN JAVA

Q1 Online Book Store Inventory Management System

50 marks

A leading online bookstore collaborates with hundreds of publishers who upload book details every day. Since publishers use different data entry formats, the uploaded information contains duplicate ISBN numbers, inconsistent book titles, unnecessary spaces, repeated author names, and multiple copies of the same book. Customers frequently search for books based on titles, authors, and publishers. The management also wants to generate category-wise inventory reports while maintaining efficient storage and retrieval of book records. The application should make extensive use of Java Collections for handling large datasets efficiently.

Develop a Java application that performs the following tasks:

- a) Read book details including ISBN, Title, Author, Publisher, Category, and Price.
- b) Use String handling methods to standardize book titles, author names, and publisher names.
- c) Store all book records in an ArrayList.
- d) Remove duplicate ISBN numbers using a HashSet.
- e) Store ISBN as the key and Book object as the value using a HashMap.
- f) Use Iterator to display books belonging to a selected category.
- g) Implement the application using Generic Collections.
- h) Generate reports showing the number of books available in each category and explain the advantages of the selected Collection classes.

Your Code

```
1 import java.util.*;
2 public class Book{
3     String isbn;
4     String title;
5     String author;
6     String publisher;
7     String category;
8     double price;
9     // Constructor
10    Book(String isbn, String title, String author,
11        String publisher, String category, double price){
12        this.isbn = isbn;
13        this.title = standardize(title);
14        this.author = standardize(author);
15        this.publisher = standardize(publisher);
16        this.category = standardize(category);
17        this.price = price;
18    }
19    /*
20     * (b) Use String handling methods to standardize book titles,
21     * author names, and publisher names.
22     */
23    static String standardize(String s) {
24        s = s.trim();
25        s = s.replaceAll("\\s+", " ");
26        String[] words = s.toLowerCase().split(" ");
```

```

27         String result = "";
28         for (String word : words) {
29             if (word.length() > 0) {
30                 result = result
31                     + Character.toUpperCase(word.charAt(0))
32                     + word.substring(1) + " ";
33             }
34         }
35         return result.trim();
36     }
37     public String toString() {
38         return "ISBN      : " + isbn +
39             "\nTitle      : " + title +
40             "\nAuthor      : " + author +
41             "\nPublisher    : " + publisher +
42             "\nCategory    : " + category +
43             "\nPrice       : " + price;
44     }
45     public static void main(String[] args) {
46         Scanner sc = new Scanner(System.in);
47         /*
48          * (a) Read book details including ISBN, Title, Author,
49          * Publisher, Category, and Price.
50          */
51         System.out.println("===== (A) READ BOOK DETAILS =====");
52         int n = sc.nextInt();
53         sc.nextLine();
54         System.out.println("Book details successfully read.");
55         /*
56          * (c) Store all book records in an ArrayList.
57          */
58         ArrayList<Book> books = new ArrayList<>();
59         /*
60          * (d) Remove duplicate ISBN numbers using a HashSet.
61          */
62         HashSet<String> isbnSet = new HashSet<>();
63         for (int i = 0; i < n; i++) {
64             String isbn = sc.nextLine().trim();
65             String title = sc.nextLine();
66             String author = sc.nextLine();
67             String publisher = sc.nextLine();
68             String category = sc.nextLine();
69             double price = sc.nextDouble();
70             sc.nextLine();
71             if (isbnSet.add(isbn)) {
72                 Book b = new Book(
73                     isbn,
74                     title,
75                     author,
76                     publisher,
77                     category,
78                     price
79                 );
80                 books.add(b);

```

```

81         } else {
82             System.out.println(
83                 "Duplicate ISBN " + isbn + " removed."
84             );
85         }
86     }
87     /*
88     * (b) String handling and standardization
89     */
90     System.out.println(
91         "\n===== (B) STRING STANDARDIZATION ====="
92     );
93     System.out.println(
94         "Title, Author, Publisher and Category standardized."
95     );
96     /*
97     * (c) ArrayList output
98     */
99     System.out.println(
100         "\n===== (C) ARRAYLIST - BOOK RECORDS ====="
101     );
102     for (Book b : books) {
103         System.out.println("\n" + b);
104     }
105     /*
106     * (d) HashSet output
107     */
108     System.out.println(
109         "\n===== (D) HASHSET - DUPLICATE ISBN ====="
110     );
111     System.out.println(
112         "Unique ISBN numbers stored: " + isbnSet.size()
113     );
114     System.out.println(
115         "Duplicate ISBN numbers are automatically rejected."
116     );
117     /*
118     * (e) Store ISBN as key and Book object as value
119     * using a HashMap.
120     */
121     HashMap<String, Book> bookMap = new HashMap<>();
122     for (Book b : books) {
123         bookMap.put(b.isbn, b);
124     }
125     System.out.println(
126         "\n===== (E) HASHMAP - ISBN AS KEY, BOOK AS VALUE ====="
127     );
128     for (Map.Entry<String, Book> entry :
129         bookMap.entrySet()) {
130
131         System.out.println(
132             "ISBN: " + entry.getKey() +
133             " -> " + entry.getValue().title
134         );

```

```

135     }
136     /*
137     * (f) Use Iterator to display books belonging
138     * to a selected category.
139     */
140     String selectedCategory = standardize(sc.nextLine());
141     System.out.println(
142         "\n===== (F) ITERATOR - SELECTED CATEGORY ====="
143     );
144     System.out.println(
145         "Selected Category: " + selectedCategory
146     );
147     Iterator<Book> iterator = books.iterator();
148     boolean found = false;
149     while (iterator.hasNext()) {
150         Book b = iterator.next();
151         if (b.category.equalsIgnoreCase(selectedCategory)) {
152             System.out.println("\n" + b);
153             found = true;
154         }
155     }
156     if (!found) {
157         System.out.println("No books found.");
158     }
159     /*
160     * (g) Implement the application using Generic Collections.
161     */

```

```

162     System.out.println(
163         "\n===== (G) GENERIC COLLECTIONS ====="
164     );
165     System.out.println("ArrayList<Book>");
166     System.out.println("HashSet<String>");
167     System.out.println("HashMap<String, Book>");
168     System.out.println("HashMap<String, Integer>");
169     System.out.println("Iterator<Book>");
170     /*
171     * (h) Generate reports showing the number of books
172     * available in each category and explain the advantages
173     * of the selected Collection classes.
174     *
175     * ANSWER:
176     *
177     * ArrayList:
178     * Stores book records and maintains insertion order.
179     *
180     * HashSet:
181     * Removes duplicate ISBN numbers and provides fast
182     * duplicate checking.
183     *
184     * HashMap:
185     * Stores ISBN as key and Book as value and provides
186     * fast retrieval using ISBN.
187     *
188     * Iterator:
189     * Traverses the collection and displays books

```

```

192         * Generic Collections:
193         * Provide type safety and avoid unnecessary casting.
194         */
195         HashMap<String, Integer> categoryCount =
196             new HashMap<>();
197         for (Book b : books) {
198             String category = b.category;
199             categoryCount.put(
200                 category,
201                 categoryCount.getDefault(category, 0) + 1
202             );
203         }
204         System.out.println(
205             "\n===== (H) CATEGORY-WISE INVENTORY REPORT ====="
206         );
207         for (Map.Entry<String, Integer> entry :
208             categoryCount.entrySet()) {
209             System.out.println(
210                 "Category: " + entry.getKey() +
211                 " | Number of Books: " +
212                 entry.getValue()
213             );
214         }
215         System.out.println(
216             "\n===== ADVANTAGES OF COLLECTION CLASSES ====="
217         );
218         System.out.println(
219             "ArrayList : Maintains insertion order and stores book records"
220         );
221         System.out.println(
222             "HashSet : Removes duplicate ISBN numbers."
223         );
224         System.out.println(
225             "HashMap : Provides fast ISBN-based book retrieval."
226         );
227         System.out.println(
228             "Iterator : Traverses book records efficiently."
229         );
230         System.out.println(
231             "Generics : Provides type safety and avoids casting."
232         );
233         sc.close();
234     }
235 }

```